Cambridge Assessment International Education

# Cambridge International AS & A Level

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| CANDIDATE NAME | |
| CENTRE NUMBER | |
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*4535261400*

**MATHEMATICS** **9709/32**

Paper 3 Pure Mathematics 3 **May/June 2021**

**1 hour 50 minutes**

You must answer on the question paper.

You will need: List of formulae (MF19)

**INSTRUCTIONS**

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- If additional space is needed, you should use the lined page at the end of this booklet; the question number or numbers must be clearly shown.
- You should use a calculator where appropriate.
- You must show all necessary working clearly; no marks will be given for unsupported answers from a calculator.
- Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place for angles in degrees, unless a different level of accuracy is specified in the question.

**INFORMATION**

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].

This document has **20** pages. Any blank pages are indicated.


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**1** Solve the inequality $|2x - 1| < 3|x + 1|$. [4]

**2** On a sketch of an Argand diagram, shade the region whose points represent complex numbers $z$ satisfying the inequalities $|z + 1 - \mathrm{i}| \leqslant 1$ and $\arg(z - 1) \leqslant \frac{3}{4}\pi$. [4]

**3** The variables $x$ and $y$ satisfy the equation $x = A(3^{-y})$, where $A$ is a constant.

**(a)** Explain why the graph of $y$ against $\ln x$ is a straight line and state the exact value of the gradient of the line. [3]

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It is given that the line intersects the $y$-axis at the point where $y = 1.3$.

**(b)** Calculate the value of $A$, giving your answer correct to 2 decimal places. [2]

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**4** Using integration by parts, find the exact value of $\int_0^2 \tan^{-1}\left(\frac{1}{2}x\right) \mathrm{d}x$. [5]

**5** The complex number $u$ is given by $u = 10 - 4\sqrt{6}\mathrm{i}$.

Find the two square roots of $u$, giving your answers in the form $a + \mathrm{i}b$, where $a$ and $b$ are real and exact. [5]

**6** **(a)** Prove that $\operatorname{cosec} 2\theta - \cot 2\theta \equiv \tan \theta$. [3]

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**(b)** Hence show that $\int_{\frac{1}{4}\pi}^{\frac{1}{3}\pi} (\operatorname{cosec} 2\theta - \cot 2\theta)\, \mathrm{d}\theta = \frac{1}{2} \ln 2$. [4]

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**7** A curve is such that the gradient at a general point with coordinates $(x, y)$ is proportional to $\frac{y}{\sqrt{x+1}}$. The curve passes through the points with coordinates $(0, 1)$ and $(3, e)$.

By setting up and solving a differential equation, find the equation of the curve, expressing $y$ in terms of $x$. [7]

**8** The equation of a curve is $y = e^{-5x} \tan^2 x$ for $-\frac{1}{2}\pi < x < \frac{1}{2}\pi$.

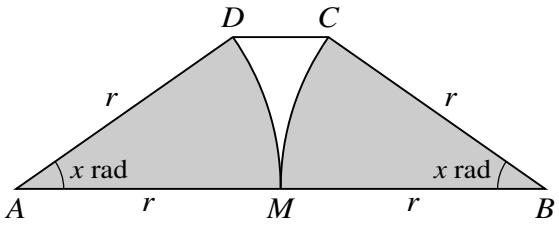
Find the $x$-coordinates of the stationary points of the curve. Give your answers correct to 3 decimal places where appropriate. [8]

**9** Let $\mathrm{f}(x) = \dfrac{14 - 3x + 2x^2}{(2 + x)(3 + x^2)}$.

**(a)** Express $\mathrm{f}(x)$ in partial fractions. [5]

**(b)** Hence obtain the expansion of $f(x)$ in ascending powers of $x$, up to and including the term in $x^2$. [5]

**10**




The diagram shows a trapezium $ABCD$ in which $AD = BC = r$ and $AB = 2r$. The acute angles $BAD$ and $ABC$ are both equal to $x$ radians. Circular arcs of radius $r$ with centres $A$ and $B$ meet at $M$, the midpoint of $AB$.

**(a)** Given that the sum of the areas of the shaded sectors is 90% of the area of the trapezium, show that $x$ satisfies the equation $x = 0.9(2 - \cos x)\sin x$. [3]

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**(b)** Verify by calculation that $x$ lies between 0.5 and 0.7. [2]

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**(c)** Show that if a sequence of values in the interval $0 < x < \frac{1}{2}\pi$ given by the iterative formula

$$x_{n+1} = \cos^{-1}\left(2 - \frac{x_n}{0.9 \sin x_n}\right)$$

converges, then it converges to the root of the equation in part **(a)**. [2]

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**(d)** Use this iterative formula to determine $x$ correct to 2 decimal places. Give the result of each iteration to 4 decimal places. [3]

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**11** With respect to the origin $O$, the points $A$ and $B$ have position vectors given by $\overrightarrow{OA} = 2\mathbf{i} - \mathbf{j}$ and $\overrightarrow{OB} = \mathbf{j} - 2\mathbf{k}$.

**(a)** Show that $OA = OB$ and use a scalar product to calculate angle $AOB$ in degrees. [4]

The midpoint of $AB$ is $M$. The point $P$ on the line through $O$ and $M$ is such that $PA : OA = \sqrt{7} : 1$.

**(b)** Find the possible position vectors of $P$. [6]

## Additional Page

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